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Q1. Choose the correct answer : (24 Marks)

1) *For transmission of normal speech signal the PCM channel needs a bandwidth of :*
   - ***a. 64KHZ***
   - ***b. 8KHz***
   - ***c. 4KHz***
2) *The capacity of an analog communication channel with 4KHz bandwidth , SNRdb = 15 db is approximately :*
   - ***a. 20000 bps***
   - ***b. 16000 bps***
   - ***c. 10000 bps***
   - ***d. 8000 bps***
3) *The core of an optical fiber has :*
   - ***a. A lower index of refraction than the air***
   - ***b. A lower index of refraction than the cladding***
   - ***c. A higher index of refraction than the cladding***
   - ***d. None Of The Above***
4) *When the index of refraction is greater in material 1 than it's in material 2 , the velocity of propgation in material 1 compared to material 2 is :*
   - ***a. Equal to OR Greater***
   - ***b. Greater***
   - ***c. Lesses***
   - ***d. Equal***

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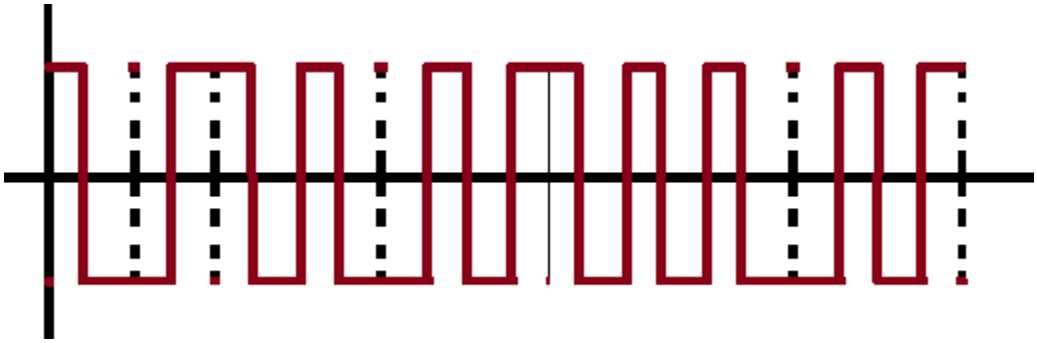
5) In the following format , how many transitions occur :

a. 16

b. 14

c. 15

d. 8



6) What is one principal defference between synchronous & Asunchronous transmission :

a. The bandwidth required a different

b. The pulse hights are different

c. The clocking is mixed with data in Asynchronous

d. The clocking is derived from the data in Synchronous

7) A example of a bounded medium is :

a. Coaxial Cable

b. Wave Guide

c. Fiber Optic Cable

d. All Of The Above

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8) *The correctness & accuracy of transmitted message content is :*

a. Verfied by the modem

b. Determine by sender & receiver not in comm. System

c. Ensured by use of digital techniques

9) *The difference between timing AND frame is :*

a. Timing is concerned with the individual bits, frame is concerned with boundries between character

b. Timing refers to serial transmission , frame refers to parallel

c. Timing is conerned primally with asynchronous system , frame is concerned with synchronous system

10) *Fiber optic cable operate at frequencies near :*

a. 20MHz

b. 200MHz

c. 2GHz

d. 800THz

11) *Transmission of binary signals requires :*

a. Less bandwidth than analog

b. More bandwidth than analog

c. The same bandwidth as analog

d. A license from FCB

12) *Codes are always :*

a. Eight bits per character

b. Either seven or eight bits per character

c. Agreed upon in advance between Tx & Rx

d. The same in all modern computers

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Q2. (10 Marks) :

If we have a screen the digitize 480x500 pixels , each pixel have a 32 different value , if we sent 30 photos per second , Find Bit Rate . (5 Marks)

Bit Rate = 480x500x32x30 = 230.4 Mbps

If this channel have bandwidth 4.5Mhz and Signal to Noise Ratio in db is 35_{db} Find capacity Of this channel . (5 Marks)

SNR_{db} = 10 Log (SNR) ➔ SNR = 3162.3

$C = 4.5M * Log_2(1+3162.3)$ ➔ 52.3Mbps

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Q3. (16 Marks) :

Reprsent (01001110) in this scheme :

a) NRZ-L
b) NRZ-I (assume the previous bit is 1)
c) AMI (assumer the previous bit 1 is represent as negative)
d) Manchester
e) Differential Manchester (assume the previous bit is 1)